

Preserving PET Quantification in Longitudinal Whole-Body Registration

Anonymous Author(s)

Anonymous Institution(s)

Abstract. In longitudinal PSMA PET/CT registration, whole-body scans are aligned that differ in field of view, body positioning, disease burden, and bowel and bladder filling, while preserving the PET-derived biomarkers that the scans provide. Alignment accuracy and preservation pull against each other: a deformation that maximises organ overlap is free to compress or expand a lesion, changing exactly the quantity a clinician reads. In this work, we adapt the diffeomorphic, coarse-to-fine LapIRN architecture to this task for the Learn2Reg 2026 PSMAReg challenge, and add two anatomically motivated constraints to its objective. The first is a set of biomarker terms that pin the metabolic tumour volume and total lesion glycolysis under the estimated transformation, including a Jacobian-based surrogate that supplies gradients throughout the lesion interior rather than only at its boundary. The second is a local rigidity constraint that promotes rigid transformation of the bones: classical derivative-based rigidity terms read a finite-difference stencil that, on ribs and cortical bone at whole-body resolution, is dominated by surrounding soft tissue, whereas a closed-form rigid fit per skeletal label reads nothing outside it. The same objective is descended per case at test time by instance optimisation. On the challenge validation cases our proposed method reaches 81.0% DSC and 6.17 mm HD95 at MTV and TLG errors of 3.3%, against 73.3%, 7.09 mm and 4.5% and 8.7% for the strongest baseline; released from the 90 s per pair the container must meet, the same method reaches 88.3% DSC, 4.80 mm HD95 and 1.4% and 2.1% biomarker error.

Keywords: Deformable image registration · PSMA PET/CT · Longitudinal imaging · Biomarker preservation.

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT has become a central imaging modality for staging, treatment planning, radiopharmaceutical therapy, and post-therapy response assessment in prostate cancer [6]. In these settings, patients often undergo multiple scans over time, comprising a baseline scan before therapy and one or more follow-up scans. Accurate baseline-to-follow-up registration can support comparison of disease burden, localisation of uptake changes, assessment of lesion progression or response, dose accumulation, and

quantitative longitudinal analysis [14]. The Learn2Reg 2026 PSMAReg challenge [9, 8, 11] formalises this problem as the estimation of a dense displacement field that warps each follow-up (moving) scan to its baseline (fixed) scan. The evaluation is performed by a combination of an organ overlap metric, boundary distance and deformation regularity. Moreover, to measure the preservation of the PET-derived tumor biomarkers, two separate metrics are used: the metabolic tumor volume (MTV) and the total lesion glycolysis (TLG).

Deformable registration has traditionally been posed as an iterative optimization over a regularised objective functional [28, 1, 24], which is accurate but slow at whole-body resolution, whereas learning-based methods amortize this optimization into network weights and replace per-pair iteration with a single forward pass at inference [3, 5, 4]. Among the latter, we build on LapIRN [20], whose Laplacian pyramid of coarse-to-fine sub-networks recovers the large deformations expected between whole-body scans while keeping the transformation smooth and invertible.

The skeleton is the main location of PSMA occurrences and thus, the accurate alignment of the bone structures is particular important. In LapIRN, global smoothness regularization is employed, which prevents local folding. However, it is desired to constrain the bone structures to move rigidly. Anatomy-aware regularisation that locally penalises non-rigid deformation of individual structures was widely explored [16, 23, 26, 19]. The regularization is therefore particularly relevant for the registration problem of the challenge.

In this work, we extend LapIRN by addressing the challenges presented above for longitudinal whole-body PSMA PET/CT registration in the PSMAReg task (Fig. 1).

Our contributions are threefold. We propose:

- an adaptation of the diffeomorphic, coarse-to-fine LapIRN architecture to longitudinal whole-body PSMA PET/CT, combining an affine pre-registration with a three-level pyramid and test-time instance optimization;
- a set of losses that constrain the PET-derived MTV and TLG given the estimated deformation, including a Jacobian determinant-based loss term;
- a local rigidity regularization for the bone structures;

2 Methods

2.1 Data and Preprocessing

We used the PSMAReg dataset of the Learn2Reg 2026 challenge [14], a longitudinal whole-body PSMA PET/CT cohort adapted from the manually annotated autoPET data of [11], in which each subject provides a baseline scan and one or more follow-up scans with a CT and a PSMA PET channel each; the follow-up is the moving and the baseline the fixed image. The images are provided already resampled to $2.7344 \times 2.7344 \times 3.27$ mm and cropped to $192 \times 192 \times 288$ voxels, so we applied no further geometric preprocessing and only normalised the

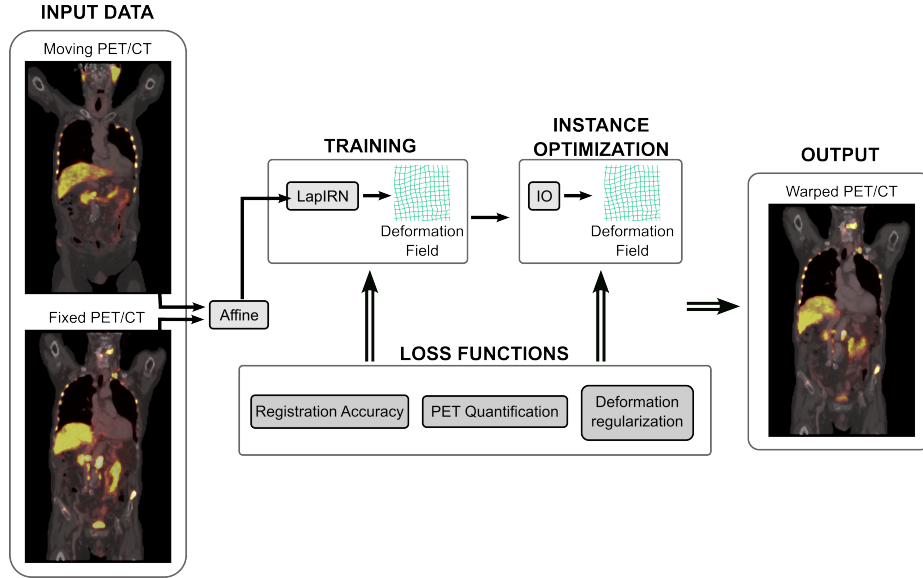


Fig. 1. Overview of the proposed registration pipeline. The moving (follow-up) and fixed (baseline) PET/CT are first coarsely aligned by an affine pre-registration (red), and the affinely aligned pair is passed to the coarse-to-fine LapIRN network, which predicts a dense displacement field (DF, green grid) used to resample the moving scan into the fixed space (warped PET/CT). Training is driven by three groups of objectives: **(1) Regularisation** on the DF, a global smoothness and folding penalty together with a per-structure bone-rigidity term, which by keeping the skeleton rigid also preserves the biomarkers of the many lesions that are intraosseous; **(2) PET preservation**, the MTV, TLG and lesion-Jacobian terms, comparing the moving and warped lesions (black dashed); and **(3) NCC + DSC**, image- and label-based similarity between the fixed and warped scans (grey dashed). At test time, the same objectives are descended per pair by instance optimisation (IO).

intensities, clipping CT to $[-1000, 1500]$ HU and the PET SUV to $[0, 20]$ and rescaling both to $[0, 1]$.

Throughout, *challenge training*, *challenge validation* and *challenge test set* denote the three splits defined by the organisers, and *our training* and *our internal validation set* the patient-wise subdivision we made of the challenge training set. From its 146 paired subjects (the 212 unpaired ones were not used) we took 117 for our training set and 29 for our internal validation set, the latter used only for checkpoint and variant selection (Sec. 2.5). Registration pairs always point from the later to the earlier time point: for training all such ordered combinations, including follow-up-to-follow-up pairs (239 pairs), and for our internal validation set every follow-up registered to its baseline (39 pairs). All results we report are obtained on the challenge validation set (20 paired subjects), as the challenge test set (~ 200 pairs of private JHU data) is held out and scored by the organisers alone.

Labels are released by the organisers for the challenge training set only: CT organ labels generated automatically with TotalSegmentator [29] and PET lesion labels manually annotated by radiologists. For the challenge validation set we therefore generated surrogate labels ourselves, with TotalSegmentator for the CT organs and the winning autoPET III submission [22] for the PET lesions; these are what both our evaluation and our instance optimisation consume on those cases.

2.2 Registration Pipeline

The proposed pipeline is described in detail below (see also Fig. 1). Throughout, I denotes the moving PET image, M its lesion mask, φ_{aff} the affine pre-registration and φ_{θ} the displacement field predicted by the network; their composition φ is the total transformation mapping the moving to the fixed image. W denotes the lesion segmentation in the fixed frame, i.e. the image of M under φ .

Step 1: Affine pre-registration and preprocessing. Baseline and follow-up scans of the same subject differ in field of view, table position and patient posture by far more than a locally regularised network can absorb, so we first remove the global component with a separate affine stage. We follow the baseline provided the challenge organisers, by removing the bed and normalizing the CTs. For each image pair we estimate an affine transform with ANTs [2] based on the CT channel alone, at half resolution and with a Mattes mutual-information metric, after masking both volumes to the body and windowing them to $[-1000, 1000]$ HU. The result is converted into a dense voxel displacement field which is then applied to the moving scan.

Step 2: Deformable registration. For the deformable stage we use LapIRN [20], a Laplacian-pyramid network whose three sub-networks operate at a quarter, a half and the full resolution of the $192 \times 192 \times 288$ volumes. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level; the sum is exponentiated by scaling and squaring to give a diffeomorphic φ_{θ} , so large deformations are recovered coarse-to-fine while the transformation stays smooth and invertible. We adapted the architecture to the multi-modal setting by extending each sub-network to take four input channels: the fixed CT and PET, and the affinely aligned moving CT and PET. Although both CT and PET are provided as network inputs, the image similarity loss is computed only on CT, since PET uptake is not conserved between time points and would otherwise drive the deformation towards matching disease rather than anatomy. PET is nevertheless used to inform the network’s prediction and to evaluate the dedicated PET preservation terms of Sec. 2.3. The levels are trained in sequence, each initialised from the previous one, which is kept frozen for the first ten epochs and fine-tuned jointly afterwards.

Step 3: Instance optimization...

2.3 Training Loss Function for deformable stage

We propose to use a weighted sum of three groups of terms, one per evaluation criterion:

$$\begin{aligned}
 \mathcal{L} = & \underbrace{\lambda_{\text{NCC}} \mathcal{L}_{\text{NCC}} + \lambda_{\text{DSC}} \mathcal{L}_{\text{DSC}}}_{\text{registration accuracy}} \\
 & + \underbrace{\lambda_{\text{MTV}} \mathcal{L}_{\text{MTV}} + \lambda_{\text{MTV-J}} \mathcal{L}_{\text{MTV-J}} + \lambda_{\text{jac}} \mathcal{L}_{\text{jac}} + \lambda_{\text{TLG}} \mathcal{L}_{\text{TLG}}}_{\text{PET quantification}} \\
 & + \underbrace{\lambda_{\text{rigid}} \mathcal{L}_{\text{rigid}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{NDV}} \mathcal{L}_{\text{NDV}}}_{\text{deformation regularisation}}.
 \end{aligned} \tag{1}$$

Each group is described below. All terms are evaluated on the composed transformation φ . Only the full-resolution level is trained with the complete objective. The two coarser levels (quarter and half resolution) use the similarity, label and regularisation terms alone, i.e. $\lambda_{\text{MTV}} = \lambda_{\text{MTV-J}} = \lambda_{\text{jac}} = \lambda_{\text{TLG}} = \lambda_{\text{rigid}} = 0$, because the lesion and bone masks they would be evaluated on are only a few voxels wide there: after downsampling, thin ribs and small lesions are largely partial-volume, so both the volume ratios and the per-structure rigid fits measure interpolation rather than anatomy.

Registration Accuracy. Image similarity is measured by the local normalised cross correlation, evaluated as a multi-resolution sum over as many scales as the level has resolutions, i.e. one scale at the coarsest and three at the full-resolution level. It is computed on the CT channel only: PET intensity is not conserved between the two time points, as lesions appear, resolve or change uptake under therapy, so a PET similarity term would reward matching disease rather than anatomy. Furthermore, we used a standard DSC loss for weak CT label supervision. The objective contains no surface-distance term, although HD95 is scored - the ablation study in Sec. 3.2 shows no benefit from including one.

Preservation of PET quantification. The MTV bias scored by the challenge is the *net* change in lesion volume under the deformation. All terms are evaluated on the composed transformation φ against the original moving lesion mask M , since the affine itself is generally not volume-preserving. The first term, $\mathcal{L}_{\text{MTV}}$, is the squared relative error between the volume of the resampled mask and that of M . It matches the scored metric exactly, but only voxels on the lesion’s sub-voxel boundary shell receive gradient from it, so the interior is free to deform as long as the boundary compensates. We therefore add a second term, $\mathcal{L}_{\text{MTV-J}}$, which pins the same net volume ratio but supplies gradients throughout the lesion interior. It rests on the change-of-variables identity: summing $\det J_\varphi$ over the warped lesion region W recovers the original lesion volume, so the mean determinant over W equals the ratio of original to warped volume, and driving that mean to one preserves it. Because only the mean is constrained, the term still permits internal redistribution of volume. Squaring the deviation per voxel

rather than after the mean gives a third, stricter term, $\mathcal{L}_{\text{jac}}$, which additionally discourages compensating compression and expansion inside the lesion.

The challenge additionally scores the TLG, the PET-activity-weighted counterpart of the MTV, which we penalize analogously as the relative change of intensity mass after warping, i.e. the absolute difference between the sums of I over M before and after resampling, divided by the former. Since both the image and the mask are resampled with the same transformation, this term is sensitive not only to volume change but also to intensity blurring at the lesion boundary, and thus complements the purely geometric terms. All four PET preservation terms are used during training and descended at test time by our instance optimisation (Sec. 2.4), so the network is trained on the objective it is later refined under.

Bone rigidity. A median of 46.0% of total lesion volume at baseline lay within skeletal structures, rising to 63.8% at follow-up. Since so much of the lesion volume is intraosseous, a rigidity penalty on a bone mask is an anatomically grounded surrogate for the biomarker terms: a rigid motion has $\det J_\varphi = 1$ and resamples the PET signal without compressing or expanding it, so it protects the lesion volume MTV measures and the intensity mass TLG measures at once.

We first used the established local rigidity penalty, which combines deviation of the Jacobian J_φ from orthogonality [16, 23] with local affineness and orientation preservation [26, 19]. Its terms are evaluated by finite differences at every voxel and only then masked, so the stencil also reads non-bone voxels and, on thin cortical bone and ribs, ends up measuring the bone-to-soft-tissue displacement discontinuity (Sec. 3.2); eroding the mask would fix this, but at whole-body resolution many bone labels are only a few voxels wide. We therefore replaced it with a per-structure rigid-fit residual over the 61 skeletal TotalSegmentator labels: for each structure we fit the best rigid transform to its own displacement in closed form by Kabsch alignment [12, 27] and penalise the mean squared residual, averaged within and then across structures. No stencil crosses the mask and each structure is fitted independently, so articulation is free, and unlike the derivative-based penalties [26, 19] it constrains only the shape of the deformation and not its placement.

Deformation regularisation. To regularise the displacement field we used a diffusion regulariser, $\mathcal{L}_{\text{smooth}}$, the mean squared spatial gradient of the displacement, together with a term penalising non-diffeomorphic voxels. Rather than the usual penalty on negative Jacobian determinants of a single finite-difference stencil, we differentially reproduce the scored metric itself: the challenge computes the non-diffeomorphic volume by decomposing each voxel into six tetrahedra and accumulating the negative part of their determinants over the body mask [15]. Writing $\det J_\varphi^{(k)}$ for the determinant of the k -th tetrahedral scheme, the loss

$$\mathcal{L}_{\text{NDV}} = \frac{100}{12|\Omega|} \sum_{p \in \Omega} \sum_{k=1}^6 \max(-\det J_\varphi^{(k)}(p), 0) \quad (2)$$

equals the scored percentage of non-diffeomorphic volume and places gradients on exactly the folded voxels, so that minimising it minimises the reported %NDV directly.

2.4 Instance Optimisation

At test time we refine each pair individually. Keeping the predicted field fixed, we optimise a residual stationary velocity field, initialised to zero and exponentiated by scaling and squaring, on top of it, so the refined transformation stays diffeomorphic by construction and the network is only ever corrected, never overwritten. We descend it for 100 Adam steps with a learning rate of 10^{-2} and keep the best rather than the last iterate, scoring each step with the DSC, HD05, MTV and TLG, so the selected step is the one the scorer would prefer.

Submission container. Instance optimisation needs CT organ segmentations of both scans and PET lesion segmentations of the moving scan, and the submitted container has to produce them and register the pair within a budget of 90 s per image pair, which the configuration above exceeds. The deployed method therefore differs from the one evaluated in validation in three points, all of them inside instance optimisation or in the labels it consumes; everything up to and including the predicted field is identical. First, PET lesion masks come from an nnU-Net [10] trained on this dataset with progressive patch-size growing [7], which is faster at inference than the ensemble used in validation. Second, CT organ labels are produced with TotalSegmentator in its fast mode together with its body segmentation. Third, instance optimisation runs until the 90s per-pair time limit is exhausted. We report both configurations separately in Sec. 3.1, since only the validation one is free of the time constraint.

2.5 Model Selection

Selection enters our pipeline twice, to keep a checkpoint within a run and to choose a variant across runs, and both use the challenge’s own ranking score, the weighted geometric mean of registration accuracy, biomarker preservation and deformation regularity with exponents 0.4, 0.4 and 0.2. Selecting on accuracy alone would be wrong in either place, because the two do not peak together: validation Dice improves until it plateaus, whereas the tumour metrics reach their optimum roughly halfway through level-3 training and degrade thereafter (Fig. 2).

2.6 Baselines

We compare against two references and two competing methods, all evaluated on the same pairs and under the identical protocol of Sec. 3.1.

As a classical iterative baseline we ran NiftyReg [18, 17], using `reg_aladin` for the affine stage followed by `reg_f3d` initialised with it, both with default

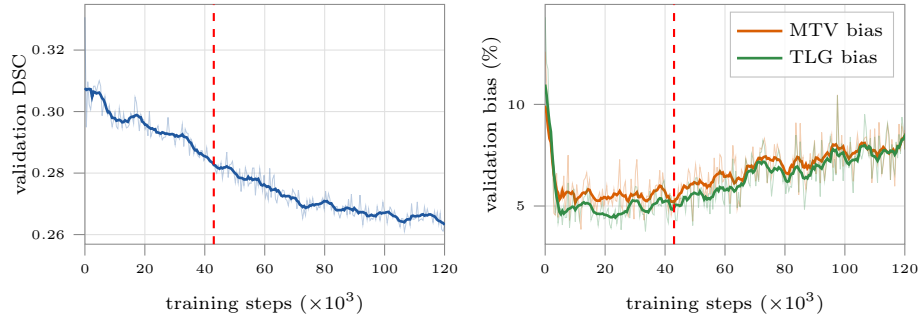


Fig. 2. Accuracy and biomarker preservation do not peak at the same checkpoint. Validation CT Dice (left) rises to a plateau, while the MTV and TLG bias (right) reach their optimum roughly halfway through level-3 training and degrade thereafter. The red dashed line marks the checkpoint selected by Eq. (??). Faint lines are the per-round values, solid lines a centred rolling mean.

parameters. As a learning-free optimisation baseline we ran ConvexAdam [25], in the form provided by the challenge organisers as the official baseline of the task.

2.7 Implementation Details

Augmentation was chosen to reproduce the ways two sessions of the same patient differ rather than to maximise variety: a left-right flip ($p = 0.5$), a CT intensity shift of ± 0.02 in the normalised window (about ± 50 HU) and scaling of 0.9 to 1.1, and a PET scaling of 0.85 to 1.15 without a shift, since SUV depends on dose, uptake time and body weight but an additive offset has no physical counterpart. We additionally crop up to 40 slices from either end of both scans along the axial axis, and then the moving and fixed scan independently by up to 10 further slices each; the latter is what reproduces a baseline/follow-up mismatch, forcing the network to register anatomy present in one scan and truncated in the other.

Each pyramid level is trained in sequence with Adam, at a batch size of one with gradients accumulated over four steps, for 60k, 60k and 120k steps at learning rates of $3 \cdot 10^{-4}$, $2 \cdot 10^{-4}$ and $2.5 \cdot 10^{-4}$. Every level starts with a five-epoch linear learning-rate warmup and keeps the preceding level frozen for its first ten epochs. The convolutional trunk runs in `bfloat16` while the transformations and all losses are kept in single precision, since scaling and squaring and the Jacobian determinants compound rounding error. The loss weights were chosen by the sweep of Sec. 3.1.

All models were trained and evaluated on a single NVIDIA H100 80GB HBM3 (driver 610.57.04), in Python 3.11 with PyTorch 2.6.0 [21] and CUDA 12.4; the affine pre-registration uses ANTsPy 0.6.3 [2], and image and array handling NiBabel 5.4.2, NumPy 2.4.4 and SciPy 1.17.1. Training all three levels takes 2d 6h 7m 6s.

3 Results

3.1 Evaluation Protocol

All numbers below are computed on the challenge validation set and, unless stated otherwise, against the surrogate labels of Sec. 2.1, since the organisers release no labels for those cases. We report our method in three configurations: *ours (container)*, the submitted method of Sec. 2.4, which is what the challenge test set will be scored on and therefore the one we test the baselines against; *ours (validation)*, the unconstrained method of Sec. 2.2, reported as the upper bound the 90s per-pair budget gives up; and *ours (no IO)*, the backbone alone, which isolates what test-time refinement contributes. The latter two carry no significance marks.

3.2 Ablation of the Objective

Derivative-based rigidity is ill posed on thin bone. We quantified the stencil leakage of Sec. 2.3 across five cases: 53.0% (range 51.6 to 54.7) of the voxels entering the loss were not bone, and 44.8% of the gradient magnitude fell outside bone. A field with zero displacement inside every bone, which is perfectly rigid by construction and should therefore cost nothing, scored 56 (range 40 to 69), all of which vanished after a single mask erosion. The term was thus measuring the bone-to-soft tissue displacement discontinuity that any correct registration must exhibit. With the per-structure rigid-fit penalty of Sec. 2.3, the zero displacement field now scores exactly 0 and a per-bone rigid field $2.4e-6$, while anisotropic squeeze is still rejected at 1.16.

A surface-distance did not increase alignment accuracy. Since HD95 is scored, a surface-distance term [13] is the obvious addition to the objective, but the 2296 scored (case, label) entries leave it nothing to do. Half the HD95 mass sits in the worst 10% of entries, and splitting it against the unregistered pair shows that 48.1% is inherent, already bad before registration and no better after, while none of it was created by our field. Those entries are limbs repositioned between sessions, hollow organs with changing filling and anatomy truncated by the field of view, none of which have a boundary to refine. The rest is already at the discretisation floor: the median per-label HD95 is one voxel and the third quartile two. We measured this before instance optimisation, so it holds for the training objective as well.

3.3 Comparison against Baselines

We swept the loss weights and the instance-optimisation settings and selected the one reported below with the challenge ranking of Sec. 2.5, applied to the surrogate protocol above. Table 1 places the three configurations against the baselines. The container configuration improves DSC from 73.3% (NiftyReg, the

Table 1. Our method against the baselines on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. 3.1. Every cell is a mean over the cases, runtime included: DSC and the MTV/TLG errors in %, NDV in ppm, and time in seconds per image pair on the hardware of Sec. 2.7. Bold marks the best value in each column. * marks metrics on which *ours (container)* is significantly better than the baseline in that row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$); runtime is not tested. Dashes mark metrics undefined for a baseline.

Method	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$	MTV %err $\downarrow$	TLG %err $\downarrow$	NDV (ppm) $\downarrow$	Time (s) $\downarrow$
Initial	14.8*	30.9*	—	—	—	—
Affine	52.9*	11.7*	5.47	4.70	—	19.1 $\pm$ 0.920
ConvexAdam	63.0*	8.41*	5.10	4.15	100	22.8 $\pm$ 1.06
NiftyReg	73.3*	7.09*	4.46	8.68*	1525*	49.7 $\pm$ 0.480
Ours (no IO)	76.5	6.65	3.22	5.08	0.00	19.5 $\pm$ 1.02
Ours (container)	81.0	6.17	3.29	3.31	864	89.8 $\pm$ 2.69
Ours (validation)	88.3	4.80	1.37	2.13	70.9	383 $\pm$ 7.77

strongest baseline) to 81.0% and reduces HD95 from 7.09 to 6.17 mm, while also reducing the MTV and TLG errors and the non-diffeomorphic volume from 1525 to 864 ppm.

3.4 Leaderboard Performance

On the validation leaderboard, our method (*Ours (validation)*) achieved a Dice score of 0.789 ± 0.0381 , HD95 of 5.99 ± 2.39 , MTV of 1.56 ± 1.22 , TLG of 4.18 ± 2.49 and NDV of $0.0041 \% \pm 0.0024\%$.

4 Discussion

The proposed method has some limitations. First, the biomarker terms are cheap to add and are descended identically during training and at test time, so the network is refined under the objective it was trained on - they help reduce the MTV and TLG bias against every baseline without costing accuracy.

Second, derivative-based rigidity penalties are not wrong in general, but they assume that a structure is thicker than the stencil used to differentiate it. On whole-body CT at this resolution, ribs and cortical bone are not, so the term reads mostly soft tissue and measures displacement discontinuity that a correct registration must produce. The per-structure rigid fit avoids this by reading nothing outside each label, and we would expect it to matter wherever rigid structures are thin relative to the voxel size, rather than as a universal replacement.

Several limitations qualify these numbers. All validation metrics are computed against automatic surrogate labels, which carry their own error; the model was selected on the same cases it is evaluated on, so the comparison against the baselines is exploratory rather than confirmatory; and no test-set results were available at the time of writing. The container configuration also loses accuracy due to its 90s per-pair budget, most of it in the segmentations that instance optimisation consumes, so a faster and more accurate segmentation stage is the most direct improvement available.

Acknowledgments.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Avants, B., Epstein, C., Grossman, M., et al.: Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain. *Medical Image Analysis* **12**(1), 26–41 (2008), special Issue on The Third International Workshop on Biomedical Image Registration - WBIR 2006
2. Avants, B.B., Tustison, N.J., Song, G., et al.: A reproducible evaluation of ants similarity metric performance in brain image registration. *NeuroImage* **54**(3), 2033–2044 (2011)
3. Balakrishnan, G., Zhao, A., Sabuncu, M.R., et al.: An unsupervised learning model for deformable medical image registration. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (June 2018)
4. Chen, J., Frey, E.C., He, Y., et al.: Transmorph: Transformer for unsupervised medical image registration. *Medical Image Analysis* **82**, 102615 (2022)
5. Dalca, A.V., Balakrishnan, G., Guttag, J., et al.: Unsupervised learning for fast probabilistic diffeomorphic registration. In: Frangi, A.F., Schnabel, J.A., Davatzikos, C., et al. (eds.) *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*. pp. 729–738. Springer International Publishing, Cham (2018)
6. Fendler, W.P., Eiber, M., Beheshti, M., et al.: 68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0. *European Journal of Nuclear Medicine and Molecular Imaging* **44**(6), 1014–1024 (Jun 2017)
7. Fischer, S.M., Kiechle, J., Daza, L., et al.: Progressive growing of patch size: Curriculum learning for accelerated and improved medical image segmentation (2025)
8. Hansen, L., Heyer, W., Großh  fmer, C., et al.: Learn2reg 2024: New benchmark datasets driving progress on new challenges. *Machine Learning for Biomedical Imaging* **3**, 775–791 (2025)
9. Hering, A., Hansen, L., Mok, T.C.W., et al.: Learn2reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning. *IEEE Transactions on Medical Imaging* **42**(3), 697–712 (2023)
10. Isensee, F., Jaeger, P.F., Kohl, S.A.A., et al.: nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods* **18**(2), 203–211 (Feb 2021)

11. Jeblick, K., Schönberg, S.O., et al.: A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions (PSMA-PET-CT-Lesions). The Cancer Imaging Archive (2024), version 1
12. Kabsch, W.: A solution for the best rotation to relate two sets of vectors. *Foundations of Crystallography* **32**(5), 922–923 (1976)
13. Karimi, D., Salcudean, S.E.: Reducing the hausdorff distance in medical image segmentation with convolutional neural networks. *IEEE Transactions on Medical Imaging* **39**(2), 499–513 (2020)
14. Learn2Reg Consortium: Learn2Reg 2026 – PSMAReg: Registration of pre-therapy and follow-up whole-body PSMA PET/CT scans. Codabench challenge (2026), <https://www.codabench.org/competitions/15724/>
15. Liu, Y., Chen, J., Wei, S., et al.: On finite difference jacobian computation in-ÅdeformableÅimage Åregistration. *International Journal of Computer Vision* **132**(9), 3678–3688 (Sep 2024)
16. Loeckx, D., Maes, F., Vandermeulen, D., et al.: Nonrigid image registration using free-form deformations with a local rigidity constraint. In: Barillot, C., Haynor, D.R., Hellier, P. (eds.) *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2004*. pp. 639–646. Springer Berlin Heidelberg, Berlin, Heidelberg (2004)
17. Modat, M., Cash, D.M., Daga, P., et al.: Global image registration using a symmetric block-matching approach. *Journal of Medical Imaging* **1**(2), 024003 (2014)
18. Modat, M., Ridgway, G.R., Taylor, Z.A., et al.: Fast free-form deformation using graphics processing units. *Computer Methods and Programs in Biomedicine* **98**(3), 278–284 (2010), hP-MICCAI 2008
19. Modersitzki, J.: Flirt with rigidity—image registration with a local non-rigidity penalty. *International Journal of Computer Vision* **76**(2), 153–163 (Feb 2008)
20. Mok, T.C.W., Chung, A.C.S.: Large deformation diffeomorphic image registration with laplacian pyramid networks. In: Martel, A.L., Abolmaesumi, P., Stoyanov, D., et al. (eds.) *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*. pp. 211–221. Springer International Publishing, Cham (2020)
21. Paszke, A., Gross, S., Massa, F., et al.: Pytorch: An imperative style, high-performance deep learning library. In: Wallach, H., Larochelle, H., Beygelzimer, A., et al. (eds.) *Advances in Neural Information Processing Systems*. vol. 32. Curran Associates, Inc. (2019)
22. Rokuss, M., Kovacs, B., Kirchhoff, Y., et al.: From fdg to psma: A hitchhiker’s guide to multitracer, multicenter lesion segmentation in pet/ct imaging. *ArXiv* (2024)
23. Ruan, D., Fessler, J.A., Roberson, M., et al.: Nonrigid registration using regularization that accommodates local tissue rigidity. In: Reinhardt, J.M., Pluim, J.P.W. (eds.) *Medical Imaging 2006: Image Processing*. vol. 6144, p. 614412. International Society for Optics and Photonics, SPIE (2006)
24. Rueckert, D., Sonoda, L., Hayes, C., et al.: Nonrigid registration using free-form deformations: application to breast mr images. *IEEE Transactions on Medical Imaging* **18**(8), 712–721 (1999)
25. Siebert, H., Hansen, L., Heinrich, M.P.: Fast 3d registration withÅaccurate optimisation andÅlittle learning forÅlearn2reg 2021. In: Aubreville, M., Zimmerer, D., Heinrich, M. (eds.) *Biomedical Image Registration, Domain Generalisation and Out-of-Distribution Analysis*. pp. 174–179. Springer International Publishing, Cham (2022)
26. Staring, M., Klein, S., Pluim, J.P.W.: A rigidity penalty term for nonrigid registration. *Medical Physics* **34**(11), 4098–4108 (2007)

27. Umeyama, S.: Least-squares estimation of transformation parameters between two point patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(4), 376–380 (1991)
28. Vercauteren, T., Pennec, X., Perchant, A., et al.: Diffeomorphic demons: Efficient non-parametric image registration. *NeuroImage* **45**(1, Supplement 1), S61–S72 (2009)
29. Wasserthal, J., Breit, H.C., Meyer, M.T., et al.: Totalsegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence* **5**(5), e230024 (2023)